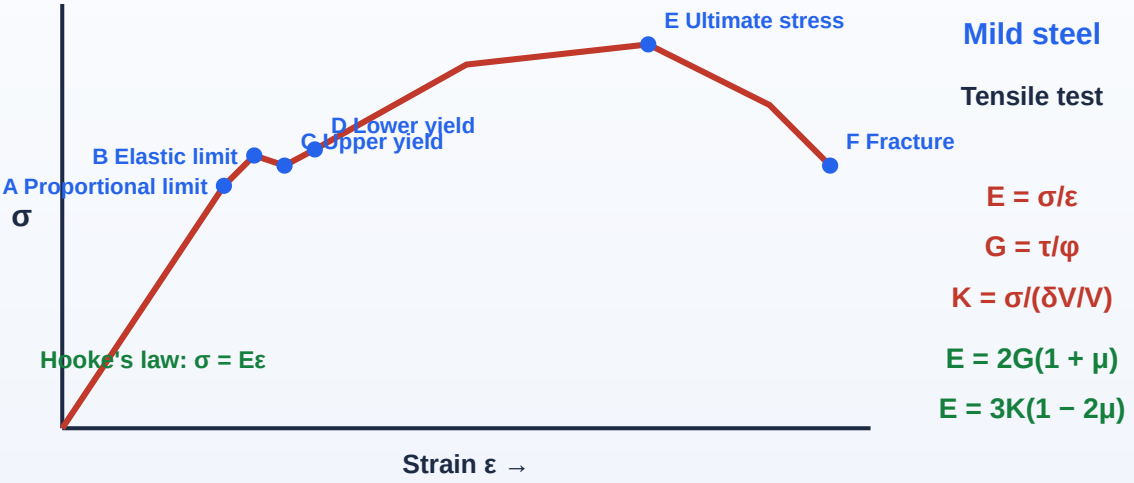
UPSSSC ALERT

# UPSSSC JE

# MECHANICAL STRENGTH OF MATERIALS

## Complete English Edition

**Unit 3 · 10 Marks · 2026**



**UPSSSC JE (Mechanical) 2026:** Part-1 has **10 questions (10 marks)** from **Strength of Materials**. Syllabus: stress and strain, elastic limit and elastic constants, riveted joints, BM and SF diagrams, stress in composite bars, torsion of circular shafts and thin-walled pressure vessels.

Each chapter has key points, diagrams, formula table, detailed explanation, solved numericals, one-liners, MCQs with explanations and SSC JE PYQs. Content follows standard BTEUP diploma textbooks (R. K. Rajput, R. S. Khurmi, S. Ramamrutham).

**For a better score, study this PDF and also attempt the QUIZ.**

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# Chapter 1: Stress, Strain & Elastic Constants



## Stress, Strain & Elastic Constants – Basic Concept

When an external force acts on a body, an internal resisting force develops. The resisting force per unit area is called **stress**, and the ratio of change in length to original length is called **strain**. Up to the **elastic limit**, the body regains its original shape when the load is removed.

## 1. Key Points

- **Stress (σ):** $\sigma = P/A$; unit N/m² (Pa) or N/mm² (MPa). Types: tensile, compressive and shear stress.
- **Strain (ε):** $\varepsilon = \delta L/L$; dimensionless. Shear strain (φ) is angular distortion; volumetric strain = $\delta V/V$.
- **Hooke's law:** up to the proportional limit, stress is proportional to strain: $\sigma = E\varepsilon$, where E is Young's modulus.
- **Elastic constants:** Young's modulus $E = \sigma/\varepsilon$, modulus of rigidity $G = \tau/\varphi$, bulk modulus $K = \sigma/(\delta V/V)$, and Poisson's ratio μ = lateral strain/longitudinal strain.
- **Relations:** $E = 2G(1 + \mu)$, $E = 3K(1 - 2\mu)$ and $E = 9KG/(3K + G)$.
- **Poisson's ratio:** 0.25–0.33 for metals; theoretical maximum 0.5 (incompressible, e.g. rubber) and nearly 0 for cork.
- **Elongation of a bar:** $\delta L = PL/(AE)$. Elongation of a bar hanging under its own weight = $\rho gL^2/(2E) = WL/(2AE)$, half that of the same load at the end.
- **Factor of safety:** ultimate (or yield) stress ÷ working (permissible) stress. Yield stress is used for ductile and ultimate stress for brittle materials.
- **Shear stress (τ):** $\tau = P/A$ (A = area of shear plane). Complementary shear: a shear stress on a plane is always accompanied by an equal shear stress on the perpendicular plane.
- **Working stress:** the maximum safe stress = yield (or ultimate) stress ÷ factor of safety.
- **Material properties:** ductility, malleability, hardness, toughness, elasticity, plasticity and brittleness.
- **Impact load:** W falling through h gives $\sigma = (W/A)[1 + \sqrt{(1 + 2AEh/(WL))}]$; with h = 0, $\sigma = 2W/A$.

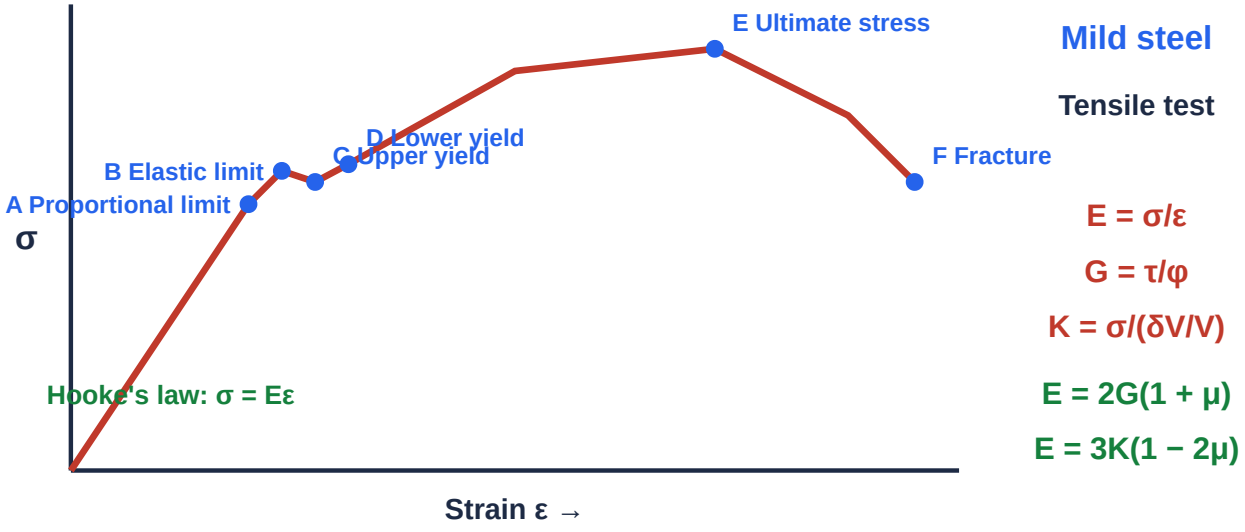



*Fig 3.1: Stress–strain diagram of mild steel and elastic constants*

## 2. Important Formulas & Facts

| Quantity / Topic | Formula / Fact |
|---|---|
| **Stress / Strain** | $\sigma = P/A$ , $\varepsilon = \delta L/L$ |
| **Hooke's law** | $\sigma = E\,\varepsilon$ |
| **Elongation** | $\delta L = PL / AE$ |
| **E–G relation** | $E = 2G(1 + \mu)$ |
| **E–K relation** | $E = 3K(1 - 2\mu)$ |
| **E, G, K relation** | $E = 9KG / (3K + G)$ |
| **Volumetric strain (bar)** | $\delta V/V = \varepsilon(1 - 2\mu)$ |
| **Elongation under own weight** | $\delta = \rho g L^2 / 2E$ |
| **Elongation of tapering bar** | $\delta = 4PL / (\pi E d_1 d_2)$ |
| **Strain energy** | $U = \sigma^2 V / 2E$ |

## 3. Detailed Explanation

- **Stress–strain diagram (mild steel):** proportional limit → elastic limit → upper and lower yield points → strain hardening → ultimate stress → necking → fracture.
- **Ductile and brittle materials:** ductile materials (mild steel, copper) show large plastic deformation and a clear yield point before fracture; brittle ones (cast iron, concrete) fail without warning.
- **Proof stress:** for materials without a distinct yield point (aluminium, high-carbon steel), yield is found by the 0.2% offset method.
- **Elastic constants:** an isotropic material has four elastic constants (E, G, K, μ), but only two are independent.

- **Strain energy and resilience:** energy stored up to elastic limit = ½ × P × δL. Maximum energy per unit volume is the modulus of resilience = $\sigma^2/2E$. A suddenly applied load produces twice the stress of a gradually applied one.
- **Percentage elongation and reduction in area:** measures of ductility in a tensile test; about 25–30% elongation for mild steel.
- **Fatigue and creep:** failure under repeated fluctuating loads even below the elastic limit is fatigue; slow increase of strain under constant load at high temperature is creep.
- **Stress concentration:** near holes, notches or sharp corners the local stress rises several times the average; it is reduced by providing fillets.

## 4. Comparison Table — Ductile vs Brittle

| Basis | Ductile | Brittle |
|---|---|---|
| Deformation before fracture | Large | Very little |
| Yield point | Distinct | None |
| Design basis | Yield stress | Ultimate stress |
| Examples | Mild steel, copper | Cast iron, glass, concrete |

## 5. Solved Numerical Problems

**Example 1.** A steel rod 2 m long and 20 mm diameter carries 31.4 kN tension. E = 200 GPa. Find stress and elongation.

**Solution:**

1. $A = \pi/4 \times 20^2 = 314.16$ mm²
2. $\sigma = 31400/314.16 =$ **100 N/mm²**
3. $\delta L = \sigma L/E = 100 \times 2000/200000 =$ **1 mm**

**Example 2.** If E = 200 GPa and μ = 0.3, find G and K.

**Solution:**

1. $G = E/2(1 + \mu) = 200/2.6 =$ **76.9 GPa**
2. $K = E/3(1 - 2\mu) = 200/1.2 =$ **166.7 GPa**

**Example 3.** Yield stress of mild steel is 250 MPa and factor of safety 2.5. Find the permissible stress.

**Solution:**

1. Permissible stress = 250/2.5 = **100 MPa**

**Example 4.** A 10 mm pin carries 15 kN in double shear. Find the shear stress.

**Solution:**

1. $\tau = P/(2 \times \pi d^2/4) = 15000/(2 \times 78.54)$
2. $\tau =$ **95.5 N/mm²**

**Example 5.** A 20 kN load is suddenly applied to a 2 m bar of area 400 mm². E = 200 GPa. Find stress and elongation.

**Solution:**

1. $\sigma = 2P/A = 2 \times 20000/400 =$ **100 N/mm²**
2. $\delta L = \sigma L/E = 100 \times 2000/200000 =$ **1 mm**

**Example 6.** A bar's length changes from 1000 to 1001.5 mm and diameter from 20 to 19.991 mm. Find Poisson's ratio.

**Solution:**

1. Longitudinal strain = 1.5/1000 = 0.0015; lateral strain = 0.009/20 = 0.00045
2. $\mu = 0.00045/0.0015 =$ **0.3**

## 6. Frequently Asked Facts

- $E = 2G(1 + \mu)$ and $E = 3K(1 - 2\mu)$ are the most asked relations.
- Maximum Poisson's ratio is 0.5; then K becomes infinite.
- Elongation under own weight is half that of the same load at the end.

## 7. Note (Confusing Facts)

- Elastic limit and proportional limit are different; Hooke's law holds only up to the proportional limit.
- Strain is dimensionless, but stress and E have the same unit (N/mm²).
- A suddenly applied load doubles the stress; an impact load gives even more.

**Remember:** One line: E > K > G for common metals ($\mu \approx 0.3$). Assertion–Reason: (A) Rubber hardly changes volume. (R) Its Poisson's ratio is about 0.5. — Both true; R explains A.

## 30 Most Important One-Liner Facts — Stress, Strain & Elastic Constants

**1. Unit of stress?**

— N/m² (Pa).

**2. Unit of strain?**

— Dimensionless.

**3. Hooke's law?**

— $\sigma = E\varepsilon$ (up to proportional limit).

**4. Unit of Young's modulus?**

— N/mm².

**5. E of steel approximately?**

— 200 GPa.

**6. Modulus of rigidity?**

— $G = \tau/\varphi$.

**7. Bulk modulus?**

— $K = \sigma/(\delta V/V)$.

**8. Poisson's ratio?**

— Lateral strain ÷ longitudinal strain.

**9. Maximum Poisson's ratio?**

— 0.5.

**10. Poisson's ratio of metals?**

— 0.25–0.33.

**11. Relation of E and G?**

— $E = 2G(1 + \mu)$.

**12. Relation of E and K?**

— $E = 3K(1 - 2\mu)$.

**13. Independent elastic constants?**

— Two.

**14. Elongation of a bar?**

— PL/AE.

**15. Elongation due to own weight?**

— $\rho gL^2/2E$.

**16. Local narrowing before fracture?**

— Necking.

**17. Design basis for ductile material?**

— Yield stress.

**18. Design basis for brittle material?**

— Ultimate stress.

**19. Stress due to sudden load?**

— Twice the gradual load stress.

**20. Modulus of resilience?**

— $\sigma^2/2E$.

**21. Shear stress?**

— $\tau = P/A$ (shear plane).

**22. Complementary shear?**

— Equal shear on perpendicular plane.

**23. Working stress?**

— Yield ÷ factor of safety.

**24. Failure under repeated loading?**

— Fatigue.

**25. Slow strain at high temperature?**

— Creep.

**26. Stress rise at sharp corners?**

— Stress concentration.

**27. Property to be drawn into wire?**

— Ductility.

**28. Property to be beaten into sheets?**

— Malleability.

**29. Resistance to fracture by absorbing energy?**

— Toughness.

**30. Percent elongation of mild steel?**

— ≈ 25–30%.

## Practice MCQs — Stress, Strain & Elastic Constants

### Q1. Hooke's law holds up to:

(a) Proportional limit
(b) Yield point
(c) Ultimate stress
(d) Fracture

✔ **Answer: (a) Proportional limit**

**Explanation:** Up to proportional limit $\sigma \propto \varepsilon$.

### Q2. Relation between E, G and μ:

(a) $E = 2G(1 + \mu)$
(b) $E = 3G(1 - 2\mu)$
(c) $E = G(1 + \mu)$
(d) $E = 2G(1 - \mu)$

✔ **Answer: (a) $E = 2G(1 + \mu)$**

**Explanation:** Standard relation.

### Q3. Maximum theoretical value of Poisson's ratio:

(a) 0.25
(b) 0.33
(c) 0.5
(d) 1

✔ **Answer: (c) 0.5**

**Explanation:** At $\mu = 0.5$ volumetric strain is zero.

### Q4. Independent elastic constants for an isotropic material:

(a) 1
(b) 2
(c) 3
(d) 4

✔ **Answer: (b) 2**

**Explanation:** Four constants, two independent.

---

### Q5. Elongation of a bar hanging under its own weight:

(a) $\rho g L^2/E$
(b) $\rho g L^2/2E$
(c) $2\rho g L^2/E$
(d) $\rho g L/2E$

✔ **Answer: (b) $\rho g L^2/2E$**

**Explanation:** Half that of the same load at the end.

---

### Q6. Factor of safety for ductile materials is based on:

(a) Ultimate stress
(b) Yield stress
(c) Breaking stress
(d) Proportional limit

✔ **Answer: (b) Yield stress**

**Explanation:** A ductile material is considered failed at yield.

---

### Q7. Stress due to a suddenly applied load is ____ that of a gradually applied load:

(a) Half
(b) Equal
(c) Twice
(d) Four times

✔ **Answer: (c) Twice**

**Explanation:** $\sigma = 2P/A$.

---

### Q8. E for mild steel is about:

(a) 70 GPa
(b) 100 GPa
(c) 200 GPa
(d) 400 GPa

✔ **Answer: (c) 200 GPa**

**Explanation:** $\approx 2 \times 10^5$ N/mm².

---

### Q9. For materials without a distinct yield point, we use:

(a) Proof stress
(b) Ultimate stress
(c) Breaking stress
(d) Poisson's ratio

✔ **Answer: (a) Proof stress**

**Explanation:** 0.2% offset method.

---

### Q10. In $E = 3K(1 - 2\mu)$, if $\mu = 0.5$, K is:

(a) Zero
(b) E
(c) Infinite
(d) E/3

✔ **Answer: (c) Infinite**

**Explanation:** The denominator becomes zero.

---

### Q11. Failure under repeated fluctuating loads below the elastic limit:

(a) Creep
(b) Fatigue
(c) Buckling
(d) Impact

✔ **Answer: (b) Fatigue**

**Explanation:** Fatigue failure.

---

### Q12. Property of a material to be drawn into wires:

(a) Malleability
(b) Ductility
(c) Hardness
(d) Brittleness

✔ **Answer: (b) Ductility**

**Explanation:** Ductility.

---

### Q13. 10 mm pin in double shear with 15 kN — τ ≈

(a) 191 MPa
(b) 95.5 MPa
(c) 47.7 MPa
(d) 150 MPa

✔ **Answer: (b) 95.5 MPa**

**Explanation:** 15000/(2 × 78.54).

---

### Q14. Slow increase of strain at high temperature under constant load:

(a) Fatigue
(b) Creep
(c) Elasticity
(d) Hardening

✔ **Answer: (b) Creep**

**Explanation:** Creep.

---

### Q15. Complementary shear stress means:

(a) Equal shear on the perpendicular plane
(b) Double shear
(c) Tensile stress
(d) Zero stress

✔ **Answer: (a) Equal shear on the perpendicular plane**

**Explanation:** Needed for equilibrium.

---

## SSC JE / UPSSSC JE Previous Year Questions — Stress, Strain & Elastic Constants

### Q1. Relation between E, K and G:

SSC JE (Previous Year)

(a) E = 9KG/(3K + G)
(b) E = 3KG/(9K + G)
(c) E = 9KG/(K + 3G)
(d) E = KG/(3K + G)

✔ **Answer: (a) E = 9KG/(3K + G)**

**Explanation:** Standard relation.

---

**Q2. In the stress–strain curve of mild steel, the first point reached is:** SSC JE (Previous Year)

(a) Elastic limit
(b) Proportional limit
(c) Yield point
(d) Ultimate point

✔ **Answer: (b) Proportional limit**

**Explanation:** Order: proportional → elastic → yield.

---

**Q3. The unit of strain is:** SSC JE (Previous Year)

(a) $N/m^2$
(b) m
(c) No unit
(d) N

✔ **Answer: (c) No unit**

**Explanation:** Ratio of lengths.

---

**Q4. If Poisson's ratio is 0.25, E/G equals:** SSC JE (Previous Year)

(a) 2.0
(b) 2.5
(c) 3.0
(d) 1.5

✔ **Answer: (b) 2.5**

**Explanation:** $E/G = 2(1 + 0.25) = 2.5$.

---

**Q5. A material that fractures without plastic deformation is:** SSC JE (Previous Year)

(a) Ductile
(b) Brittle
(c) Malleable
(d) Tough

✔ **Answer: (b) Brittle**

**Explanation:** Brittle.

---

**Q6. Necking in the stress–strain curve begins at:** SSC JE (Previous Year)

(a) Elastic limit
(b) Yield point
(c) Ultimate stress point
(d) Fracture point

✔ **Answer: (c) Ultimate stress point**

**Explanation:** Necking starts after ultimate stress.

---

# Chapter 2: Riveted Joints



## Riveted Joints – Basic Concept

A **rivet** is a permanent fastener used to join plates in boilers, tanks, bridges and ships. It has a head, shank and tail; it is inserted hot and the second head is formed by hammering.

## 1. Key Points

- **Types of joints:** lap joint (plates overlap) and butt joint (plates end to end with one or two cover plates).
- **Rows of rivets:** single, double or triple riveted; in multiple rows, chain or zig-zag arrangement.
- **Pitch (p):** distance between centres of adjacent rivets in a row. **Margin (m):** distance from rivet centre to plate edge ≈ 1.5d.
- **Unwin's formula:** rivet hole diameter $d = 6\sqrt{t}$ (mm), where t is plate thickness in mm.
- **Modes of failure:** (i) tearing of plate: $P_t = (p - d)t\sigma_t$; (ii) shearing of rivet: $P_s = n(\pi/4)d^2\tau$; (iii) crushing: $P_c = n \cdot d \cdot t \cdot \sigma_c$.
- **Shear:** in a lap joint and single-cover butt joint the rivet is in single shear; in a double-cover butt joint it is in double shear (strength taken as 1.875 times, IBR).
- **Efficiency of joint:** $\eta$ = least of $P_t$, $P_s$, $P_c$ ÷ strength of solid plate ($p \cdot t \cdot \sigma_t$). A riveted joint is never 100% efficient.
- **Caulking and fullering:** plate edges are pressed with a caulking tool to make the joint leak-proof; fullering gives a better seal.
- **Pitch values:** minimum pitch ≈ 2.5d–3d (room for hammering); maximum pitch limited by leak-tightness and buckling.
- **Back pitch:** distance between two consecutive rows; **diagonal pitch:** diagonal distance between rivets of adjacent rows in zig-zag riveting.
- **Welded vs riveted:** welding is lighter, leak-proof and more efficient (≈ 100%); riveting suits dissimilar metals and vibrating structures.

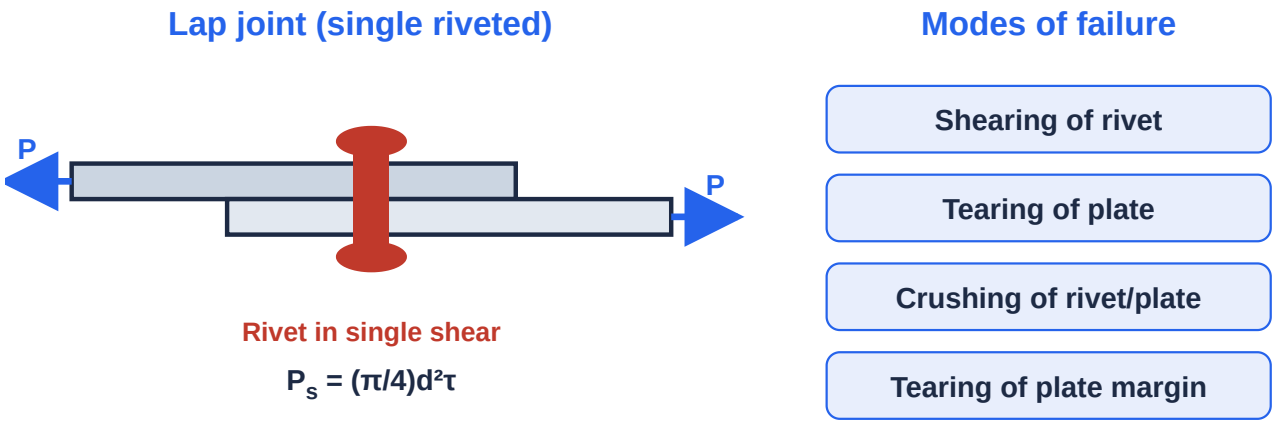



*Fig 3.2: Riveted lap joint and modes of failure*

## 2. Important Formulas & Facts

| Quantity / Topic | Formula / Fact |
|---|---|
| **Unwin's formula** | $d = 6\sqrt{t}$ (mm) |
| **Tearing of plate** | $P_t = (p - d)\, t\, \sigma_t$ |
| **Shearing of rivet (single)** | $P_s = n\,(\pi/4)\, d^2\, \tau$ |
| **Shearing of rivet (double)** | $P_s = 2n\,(\pi/4)\, d^2\, \tau$ |
| **Crushing** | $P_c = n\, d\, t\, \sigma_c$ |
| **Strength of solid plate** | $P = p\, t\, \sigma_t$ |
| **Efficiency** | $\eta = \min(P_t, P_s, P_c) / (p\, t\, \sigma_t)$ |
| **Tearing efficiency** | $\eta_t = (p - d)/p$ |

## 3. Detailed Explanation

- **Best design:** a joint is most balanced when its tearing, shearing and crushing strengths are nearly equal.
- **Joints in boilers:** lap joints are used for circumferential seams and butt joints for longitudinal seams, because the hoop stress is double.
- **Types of rivet heads:** snap (button) head is the most common; countersunk head where a flush surface is needed; pan and conical heads are also used.
- **Rivet material:** ductile — mild steel, wrought iron, copper, aluminium — so that the head can be formed by hammering.
- **Double-riveted joint calculation:** with n = 2 rivets per pitch, $P_s = 2(\pi/4)d^2\tau$ and $P_c = 2dt\sigma_c$; tearing only along the outer row $(p - d)t\sigma_t$.
- **Ways to increase efficiency:** more rows, double cover plates, diamond riveting (fewer rivets in the outer row) — these raise the tearing strength.

## 4. Comparison Table — Lap Joint vs Butt Joint

| Basis | Lap joint | Butt joint |
|---|---|---|
| Plate position | Overlapping | End to end + cover plate |
| Rivet shear | Single | Double (with two covers) |
| Strength/efficiency | Lower | Higher |
| Use in boilers | Circumferential seam | Longitudinal seam |

## 5. Solved Numerical Problems

**Example 1.** Find the rivet diameter for a 16 mm plate by Unwin's formula.

**Solution:**

1. $d = 6\sqrt{16} = 6 \times 4 =$ **24 mm**

**Example 2.** Single riveted lap joint: p = 60 mm, d = 20 mm, t = 10 mm, $\sigma_t = 100$, $\tau = 80$, $\sigma_c = 160$ N/mm² (one rivet per pitch). Find the efficiency.

**Solution:**

1. $P_t = (60 - 20) \times 10 \times 100 = 40{,}000$ N
2. $P_s = (\pi/4) \times 400 \times 80 = 25{,}133$ N
3. $P_c = 20 \times 10 \times 160 = 32{,}000$ N
4. Solid plate $= 60 \times 10 \times 100 = 60{,}000$ N
5. $\eta = 25{,}133/60{,}000 =$ **41.9%**

**Example 3.** Double-riveted lap joint: p = 80, d = 20, t = 10 mm, $\sigma_t = 100$, $\tau = 75$, $\sigma_c = 150$ N/mm². Find the efficiency.

**Solution:**

1. $P_t = (80 - 20) \times 10 \times 100 = 60{,}000$ N
2. $P_s = 2 \times (\pi/4) \times 400 \times 75 = 47{,}124$ N
3. $P_c = 2 \times 20 \times 10 \times 150 = 60{,}000$ N
4. Solid $= 80 \times 10 \times 100 = 80{,}000$ N
5. $\eta = 47{,}124/80{,}000 =$ **58.9%**

**Example 4.** Find Unwin's diameter and minimum pitch (3d) for a 25 mm plate.

**Solution:**

1. $d = 6\sqrt{25} =$ **30 mm**
2. Minimum pitch $= 3 \times 30 =$ **90 mm**

## 6. Frequently Asked Facts

- Unwin's formula $d = 6\sqrt{t}$ is frequently asked.
- Efficiency = least strength ÷ strength of solid plate.

- Butt joint for longitudinal seam of a boiler.

## 7. Note (Confusing Facts)

- Caulking makes a joint leak-proof, not stronger.
- Tearing uses (p − d) because the hole reduces the plate area.
- Crushing area is d × t (projected area), not $\pi d^2/4$.

**Remember:** One line: $d = 6\sqrt{t}$; $\eta = (p - d)/p$ for tearing. Assertion–Reason: (A) The longitudinal seam of a boiler is made stronger. (R) Hoop stress is twice the longitudinal stress. — Both true; R explains A.

## 30 Most Important One-Liner Facts — Riveted Joints

**1. Rivet is which fastener?**

— Permanent.

**2. Unwin's formula?**

— $d = 6\sqrt{t}$.

**3. Pitch?**

— Distance between adjacent rivet centres.

**4. Margin?**

— ≈ 1.5d.

**5. Overlapping plates joint?**

— Lap joint.

**6. Plates end to end?**

— Butt joint.

**7. Rivet shear in lap joint?**

— Single.

**8. Double-cover butt joint?**

— Double shear.

**9. Tearing formula?**

— $(p - d)t\sigma_t$.

**10. Crushing formula?**

— $n\ d\ t\ \sigma_c$.

**11. Strength of solid plate?**

— $p\ t\ \sigma_t$.

**12. Efficiency of joint?**

— Least strength ÷ solid plate.

**13. Process to make leak-proof?**

— Caulking / fullering.

**14. Most common rivet head?**

— Snap (button) head.

**15. Head for flush surface?**

— Countersunk.

**16. Property of rivet material?**

— Ductility.

**17. Boiler longitudinal seam?**

— Butt joint.

**18. Boiler circumferential seam?**

— Lap joint.

**19. Can efficiency be 100%?**

— No.

**20. Efficiency in tearing?**

— (p − d)/p.

**21. Minimum pitch?**

— ≈ 2.5d–3d.

**22. Distance between rows?**

— Back pitch.

**23. Diagonal distance in zig-zag?**

— Diagonal pitch.

**24. Efficiency of welded joint?**

— ≈ 100%.

**25. Joining dissimilar metals?**

— Riveting.

**26. Benefit of diamond riveting?**

— Higher efficiency.

**27. Rivet inserted in which state?**

— Red hot.

**28. Parts of rivet?**

— Head, shank, tail.

**29. Rivets per pitch in double riveting?**

— 2.

**30. Double shear factor (IBR)?**

— 1.875.

## Practice MCQs — Riveted Joints

### Q1. Rivet diameter by Unwin's formula:

(a) $d = 6\sqrt{t}$

(b) $d = 1.5t$

(c) $d = \sqrt{(6t)}$

(d) $d = 6t$

✔ **Answer: (a) $d = 6\sqrt{t}$**

**Explanation:** $d = 6\sqrt{t}$ (mm).

---

### Q2. In a lap joint the rivet is in:

(a) Single shear

(b) Double shear

(c) Torsion

(d) Bending

✔ **Answer: (a) Single shear**

**Explanation:** One shear plane between two plates.

---

### Q3. Tearing strength of plate:

(a) $(p - d)t\sigma_t$

(b) $p\ t\ \sigma_t$

(c) $d\ t\ \sigma_c$

(d) $(\pi/4)d^2\tau$

✔ **Answer: (a) $(p - d)t\sigma_t$**

**Explanation:** Area between holes.

---

### Q4. To make a riveted joint leak-proof:

(a) Caulking

(b) Drilling

(c) Reaming

(d) Punching

✔ **Answer: (a) Caulking**

**Explanation:** Edges pressed to seal.

---

### Q5. Rivet diameter for 16 mm plate (Unwin):

(a) 16 mm

(b) 20 mm

(c) 24 mm

(d) 32 mm

✔ **Answer: (c) 24 mm**

**Explanation:** $6\sqrt{16} = 24$.

---

### Q6. Crushing area of a rivet:

(a) $\pi d^2/4$

(b) $d \times t$

(c) $p \times t$

(d) $(p - d) \times t$

✔ **Answer: (b) $d \times t$**

**Explanation:** Projected area.

---

### Q7. In a double-cover butt joint the rivet is in:

(a) Single shear
(b) Double shear
(c) Crushing only
(d) Tension

✔ **Answer: (b) Double shear**

**Explanation:** Two shear planes.

---

### Q8. Efficiency of a joint is based on:

(a) Maximum strength
(b) Least strength
(c) Average strength
(d) Crushing strength only

✔ **Answer: (b) Least strength**

**Explanation:** A joint fails in its weakest mode.

---

### Q9. p = 50 mm, d = 20 mm — tearing efficiency:

(a) 40%
(b) 60%
(c) 70%
(d) 80%

✔ **Answer: (b) 60%**

**Explanation:** (50 − 20)/50 = 0.6.

---

### Q10. Joint used for the longitudinal seam of a boiler:

(a) Lap
(b) Butt
(c) Any
(d) None

✔ **Answer: (b) Butt**

**Explanation:** Stronger butt joint for higher hoop stress.

---

### Q11. Distance between two rows of rivets is called:

(a) Pitch
(b) Back pitch
(c) Margin
(d) Diagonal pitch

✔ **Answer: (b) Back pitch**

**Explanation:** Back pitch.

---

### Q12. Minimum pitch of rivets is generally:

(a) d
(b) 1.5d
(c) 2.5d–3d
(d) 6d

✔ **Answer: (c) 2.5d–3d**

**Explanation:** Room for hammering.

---

### Q13. Compared to a welded joint, a riveted joint is:

(a) More efficient
(b) Less efficient
(c) Lighter
(d) More leak-proof

✔ **Answer: (b) Less efficient**

**Explanation:** Holes weaken it.

---

### Q14. Rivet material should be:

(a) Brittle
(b) Ductile
(c) Hard
(d) Cast

✔ **Answer: (b) Ductile**

**Explanation:** To form the head.

---

### Q15. Rivets per pitch length in a double-riveted lap joint:

(a) 1
(b) 2
(c) 3
(d) 4

✔ **Answer: (b) 2**

**Explanation:** Two rows.

---

## SSC JE / UPSSSC JE Previous Year Questions — Riveted Joints

### Q1. Efficiency of a riveted joint is:

SSC JE (Previous Year)

(a) More than 100%
(b) Always less than 100%
(c) 100%
(d) Zero

✔ **Answer: (b) Always less than 100%**

**Explanation:** Holes weaken the plate.

---

### Q2. Margin of a rivet is generally:

SSC JE (Previous Year)

(a) 0.5d
(b) d
(c) 1.5d
(d) 3d

✔ **Answer: (c) 1.5d**

**Explanation:** m ≈ 1.5d.

---

### Q3. Purpose of caulking:

SSC JE (Previous Year)

(a) Strengthen the joint
(b) Make joint leak-proof
(c) Remove rivets
(d) Prevent rust

✔ **Answer: (b) Make joint leak-proof**

**Explanation:** Leak-proof.

---

### Q4. Rivets for boiler plates are generally:

SSC JE (Previous Year)

(a) Snap head
(b) Pan head
(c) Countersunk
(d) Flat head

✔ **Answer: (a) Snap head**

**Explanation:** Snap/button head is common.

---

**Q5. Rivet hole diameter compared to rivet diameter:**

SSC JE (Previous Year)

(a) Slightly larger

(b) Slightly smaller

(c) Equal

(d) Double

✔ **Answer: (a) Slightly larger**

**Explanation:** ≈ 1.5 mm larger.

UPSSSC ALERT

# Chapter 3: Shear Force & Bending Moment Diagrams



## Shear Force & Bending Moment Diagrams – Basic Concept

A **beam** is a structural member loaded perpendicular to its axis. At a section, the algebraic sum of vertical forces on one side is the **shear force**, and the sum of their moments is the **bending moment**. Their plots are the SFD and BMD.

## 1. Key Points

- **Types of beams:** cantilever (one end fixed), simply supported, overhanging, fixed and continuous beams.
- **Types of loads:** point load, uniformly distributed load (UDL, w N/m) and uniformly varying load (UVL, triangular).
- **Sign convention:** upward force on the left gives positive SF; sagging BM (concave up) is positive and hogging is negative.
- **Relation of load, SF and BM:** $dF/dx = -w$ and $dM/dx = F$. Hence BM is maximum where SF is zero or changes sign.
- **Shape of diagrams:** point load → SFD rectangular (horizontal), BMD linear; UDL → SFD linear, BMD parabolic; UVL → SFD parabolic, BMD cubic.
- **Point of contraflexure:** the point where BM changes sign (BM = 0); it occurs in overhanging beams.
- **In a cantilever the maximum BM** is always at the fixed end; in a simply supported beam it is usually near mid-span.
- **Overhanging beam:** one or both ends extend beyond the supports; it has both positive and negative BM and a point of contraflexure.
- **SS beam with UVL (0 to w):** $R_A = wL/6$, $R_B = wL/3$; $M_{max} = wL^2/(9\sqrt{3}) \approx 0.064wL^2$ at $x = L/\sqrt{3}$.
- **Bending equation:** $M/I = \sigma/y = E/R$; section modulus $Z = I/y_{max}$; rectangle $Z = bd^2/6$, circle $Z = \pi d^3/32$.

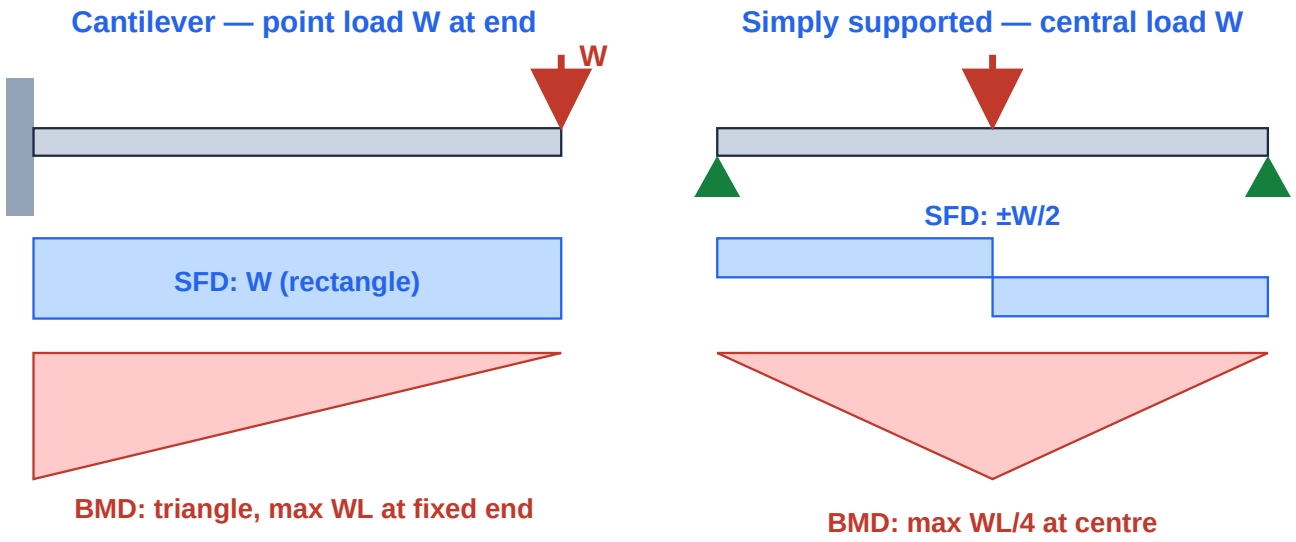



*Fig 3.3: SF and BM diagrams of a cantilever and a simply supported beam*

## 2. Important Formulas & Facts

| Quantity / Topic | Formula / Fact |
|---|---|
| **Cantilever – W at free end** | **$SF_{max} = W$ , $M_{max} = WL$** |
| **Cantilever – UDL over span** | **$SF_{max} = wL$ , $M_{max} = wL^2/2$** |
| **SS beam – central W** | **$R = W/2$ , $M_{max} = WL/4$** |
| **SS beam – UDL** | **$R = wL/2$ , $M_{max} = wL^2/8$** |
| **SS beam – W at a, b** | **$M_{max} = Wab/L$** |
| **Cantilever – UVL (0 to w)** | **$M_{max} = wL^2/6$** |
| **Relations** | **$dM/dx = F$ , $dF/dx = -w$** |

## 3. Detailed Explanation

- **Drawing the SFD:** first find reactions from ΣFy = 0 and ΣM = 0; then move from left to right showing a jump in SF at each load.
- **Drawing the BMD:** BM at a section = moment of forces on one side. The slope of the BMD equals the SF, so BM is obtained from the area of the SFD.
- **Couple (moment) load:** a couple at a point causes a sudden jump in the BMD but has no effect on the SFD.
- **Pure bending:** the portion of a beam where SF = 0 and BM is constant, e.g. between two symmetrical point loads.
- **Neutral axis:** the layer with zero bending stress; it passes through the centroid. Compression above and tension below in sagging.
- **Shear stress in beams:** $\tau = FA\bar{y}/(Ib)$; for a rectangle $\tau_{max} = 1.5\tau_{avg}$ and for a circle $4/3\ \tau_{avg}$.
- **Beam of uniform strength:** a beam in which the maximum bending stress is the same at every section, e.g. a leaf spring.

## 4. Comparison Table — Cantilever vs Simply Supported

| Basis | Cantilever | Simply supported |
|---|---|---|
| Supports | One end fixed | Both ends simply supported |
| $BM_{max}$ (point load) | WL (at fixed end) | WL/4 (at centre) |
| $BM_{max}$ (UDL) | $wL^2/2$ | $wL^2/8$ |
| Nature of BM | Hogging (negative) | Sagging (positive) |

## 5. Solved Numerical Problems

**Example 1.** A 4 m simply supported beam carries a UDL of 10 kN/m. Find the reactions and maximum BM.

**Solution:**

1. $R_A = R_B = wL/2 = 10 \times 4/2 =$ **20 kN**
2. $M_{max} = wL^2/8 = 10 \times 16/8 =$ **20 kN·m** (at centre)

**Example 2.** A 3 m cantilever carries 5 kN at its free end. Find maximum SF and BM.

**Solution:**

1. $SF_{max} =$ **5 kN**
2. $M_{max} = WL = 5 \times 3 =$ **15 kN·m** (at the fixed end)

**Example 3.** A 6 m SS beam has 30 kN at 2 m from A. Find the maximum BM.

**Solution:**

1. $M_{max} = Wab/L = 30 \times 2 \times 4/6 =$ **40 kN·m**

**Example 4.** A 2 m cantilever carries 5 kN/m UDL and 10 kN at the free end. Find maximum SF and BM.

**Solution:**

1. $SF_{max} = 10 + 5 \times 2 =$ **20 kN**
2. $M_{max} = 10 \times 2 + 5 \times 2^2/2 =$ **30 kN·m**

**Example 5.** A 200 × 300 mm rectangular beam has M = 30 kN·m. Find the maximum bending stress.

**Solution:**

1. $Z = bd^2/6 = 200 \times 300^2/6 = 3 \times 10^6\ mm^3$
2. $\sigma = M/Z = 30 \times 10^6/3 \times 10^6 =$ **10 N/mm²**

## 6. Frequently Asked Facts

- SS beam with UDL: $M_{max} = wL^2/8$; central point load: WL/4.
- BM is maximum where SF = 0.
- Under UDL the SFD is linear and BMD parabolic.

## 7. Note (Confusing Facts)

- A point load causes a jump in the SFD but only a change of slope in the BMD.
- A couple load does not affect the SFD.
- BM in a cantilever is negative (hogging).

**Remember:** One line: dM/dx = F, dF/dx = −w. Assertion–Reason: (A) An overhanging beam has a point of contraflexure. (R) BM changes sign there. — Both true; R explains A.

## 30 Most Important One-Liner Facts — Shear Force & Bending Moment Diagrams

**1. Beam with one end fixed?**

— Cantilever.

**2. UDL stands for?**

— Uniformly Distributed Load.

**3. Sign of sagging BM?**

— Positive.

**4. Sign of hogging BM?**

— Negative.

**5. dM/dx equals?**

— Shear force.

**6. Where is BM maximum?**

— Where SF = 0.

**7. SFD under point load?**

— Rectangular (horizontal).

**8. BMD under point load?**

— Linear (triangular).

**9. SFD under UDL?**

— Linear (inclined).

**10. BMD under UDL?**

— Parabolic.

**11. BMD under UVL?**

— Cubic.

**12. Cantilever, W at end — $M_{max}$?**

— WL.

**13. Cantilever, UDL — $M_{max}$?**

— $wL^2/2$.

**14. SS, central W — $M_{max}$?**

— WL/4.

**15. SS, UDL — $M_{max}$?**

— $wL^2/8$.

**16. Point where BM changes sign?**

— Point of contraflexure.

**17. Effect of couple on SFD?**

— None.

**18. Location of $M_{max}$ in cantilever?**

— At fixed end.

**19. SF in pure bending?**

— Zero.

**20. SS beam, UDL — reaction?**

— wL/2.

**21. Bending equation?**

— $M/I = \sigma/y = E/R$.

**22. Section modulus of rectangle?**

— $bd^2/6$.

**23. Section modulus of circle?**

— $\pi d^3/32$.

**24. Bending stress at neutral axis?**

— Zero.

**25. Neutral axis passes through?**

— Centroid.

**26. $\tau_{max}/\tau_{avg}$ for rectangle?**

— 1.5.

**27. $\tau_{max}/\tau_{avg}$ for circle?**

— 4/3.

**28. Example of uniform strength beam?**

— Leaf spring.

**29. Special point in overhanging beam?**

— Point of contraflexure.

**30. SS beam, UVL — $M_{max}$?**

— $wL^2/(9\sqrt{3})$.

## Practice MCQs — Shear Force & Bending Moment Diagrams

### Q1. For an SS beam with UDL over the whole span, $M_{max}$ =

(a) $wL^2/2$

(b) $wL^2/4$

(c) $wL^2/8$

(d) $wL^2/12$

✔ **Answer: (c) $wL^2/8$**

**Explanation:** $wL^2/8$ at the centre.

---

### Q2. For a cantilever with W at the free end, $M_{max}$ =

(a) WL/4

(b) WL/2

(c) WL

(d) 2WL

✔ **Answer: (c) WL**

**Explanation:** WL at the fixed end.

---

### Q3. Bending moment is maximum where:

(a) SF is maximum

(b) SF is zero

(c) Load is maximum

(d) Deflection is zero

✔ **Answer: (b) SF is zero**

**Explanation:** dM/dx = F = 0.

---

### Q4. Shape of BMD under UDL:

(a) Rectangular

(b) Triangular

(c) Parabolic

(d) Cubic

✔ **Answer: (c) Parabolic**

**Explanation:** $M \propto x^2$.

---

### Q5. SFD under point loads is:

(a) Horizontal lines

(b) Inclined lines

(c) Parabola

(d) Circle

✔ **Answer: (a) Horizontal lines**

**Explanation:** SF is constant between loads.

---

### Q6. At a point of contraflexure:

(a) SF = 0

(b) BM = 0

(c) BM is maximum

(d) Deflection is maximum

✔ **Answer: (b) BM = 0**

**Explanation:** BM changes sign.

---

**Q7. For a cantilever with UDL, $M_{max}$ =**

(a) $wL^2/2$ (b) $wL^2/8$

(c) $wL^2/6$ (d) wL

✔ **Answer: (a) $wL^2/2$**

**Explanation:** $wL^2/2$ at the fixed end.

---

**Q8. dM/dx equals:**

(a) Load intensity (b) Shear force

(c) Deflection (d) Slope

✔ **Answer: (b) Shear force**

**Explanation:** Slope of BM = SF.

---

**Q9. A couple applied at a point affects:**

(a) Only SFD (b) Only BMD

(c) Both (d) Neither

✔ **Answer: (b) Only BMD**

**Explanation:** Jump in BMD; SFD unaffected.

---

**Q10. 6 m SS beam with 20 kN at centre — $M_{max}$:**

(a) 20 kN·m (b) 30 kN·m

(c) 60 kN·m (d) 120 kN·m

✔ **Answer: (b) 30 kN·m**

**Explanation:** $WL/4 = 20 \times 6/4 = 30$.

---

**Q11. The bending equation is:**

(a) $M/I = \sigma/y = E/R$ (b) $T/J = \tau/R = G\theta/L$

(c) $M/Z = E/R$ (d) $\sigma = P/A$

✔ **Answer: (a) $M/I = \sigma/y = E/R$**

**Explanation:** Standard.

---

**Q12. Section modulus of a rectangular section:**

(a) $bd^3/12$ (b) $bd^2/6$

(c) $bd^2/12$ (d) $bd/6$

✔ **Answer: (b) $bd^2/6$**

**Explanation:** $I/y = (bd^3/12)/(d/2)$.

---

**Q13. Bending stress at the neutral axis:**

(a) Maximum (b) Zero

(c) Average (d) Infinite

✔ **Answer: (b) Zero**

**Explanation:** y = 0.

---

**Q14. Max shear stress in a rectangular beam is ____ the average:**

(a) 1.0 (b) 4/3

(c) 1.5 (d) 2.0

✔ **Answer: (c) 1.5**

**Explanation:** 1.5 times.

---

**Q15. Cantilever 2 m, UDL 5 kN/m + 10 kN at end — $M_{max}$:**

(a) 20 kN·m (b) 30 kN·m

(c) 40 kN·m (d) 10 kN·m

✔ **Answer: (b) 30 kN·m**

**Explanation:** 20 + 10 = 30.

---

## SSC JE / UPSSSC JE Previous Year Questions — Shear Force & Bending Moment Diagrams

**Q1. Maximum BM of an SS beam with central point load W:** SSC JE (Previous Year)

(a) WL/2 (b) WL/4

(c) WL/8 (d) WL

✔ **Answer: (b) WL/4**

**Explanation:** M = WL/4.

---

**Q2. A portion of a beam with zero SF and constant BM is in:** SSC JE (Previous Year)

(a) Pure bending (b) Pure shear

(c) Torsion (d) Buckling

✔ **Answer: (a) Pure bending**

**Explanation:** Pure bending.

---

**Q3. Under UVL the SFD is:** SSC JE (Previous Year)

(a) Linear (b) Parabolic

(c) Cubic (d) Rectangular

✔ **Answer: (b) Parabolic**

**Explanation:** SF ∝ $x^2$.

---

**Q4. Maximum BM in a cantilever occurs at:** SSC JE (Previous Year)

(a) Free end (b) Middle

(c) Fixed end (d) Anywhere

✔ **Answer: (c) Fixed end**

**Explanation:** At the fixed end.

---

**Q5. Maximum BM of an SS beam under UDL:** SSC JE (Previous Year)

(a) $wL^2/2$ (b) $wL^2/8$

(c) $wL^2/12$ (d) $wL^2/4$

✔ **Answer: (b) $wL^2/8$**

**Explanation:** $wL^2/8$.

---

**Q6. Under sagging BM the top fibres of a beam are in:** SSC JE (Previous Year)

(a) Tension (b) Compression

(c) Shear (d) Zero stress

✔ **Answer: (b) Compression**

**Explanation:** Compression at top, tension at bottom.

---

UPSSSC ALERT

# Chapter 4: Composite Bars & Thermal Stresses



## Composite Bars & Thermal Stresses – Basic Concept

When bars of two or more materials are joined so that all deform equally under load, the arrangement is a **composite bar**. If the expansion of a bar due to temperature change is prevented, **thermal stress** develops in it.

## 1. Key Points

- **Conditions for a composite bar:** (i) total load = sum of loads carried: $P = \sigma_1 A_1 + \sigma_2 A_2$; (ii) equal strain: $\sigma_1/E_1 = \sigma_2/E_2$.
- **Modular ratio:** $m = E_1/E_2$, so $\sigma_1 = m \cdot \sigma_2$. The material with higher E carries higher stress.
- **Thermal strain:** $\varepsilon_t = \alpha \Delta T$; free expansion $\delta = L\alpha\Delta T$.
- **Thermal stress (fully restrained):** $\sigma = E\alpha\Delta T$ — independent of length and cross-section. Heating gives compression; cooling gives tension.
- **Partial restraint:** if a gap $\delta_g$ is allowed, $\sigma = E(L\alpha\Delta T - \delta_g)/L$.
- **Thermal stress in composite bars:** on heating, the material with higher α (copper) goes into compression and the lower-α one (steel) into tension; force balance $\sigma_1 A_1 = \sigma_2 A_2$.
- **Coefficient of linear expansion (α):** steel ≈ $12 \times 10^{-6}$ /°C, copper ≈ $17 \times 10^{-6}$ /°C, aluminium ≈ $23 \times 10^{-6}$ /°C.
- **Use of modular ratio:** a composite bar is converted to one equivalent material with $A_{eq} = A_1 + mA_2$.
- **Effect of α:** for the same ΔT the higher-α material wants to expand more; when restrained it develops more stress.
- **Pipes and bolts:** bolts of a hot pipe flange go into tension on cooling; long welded rails need stress control.

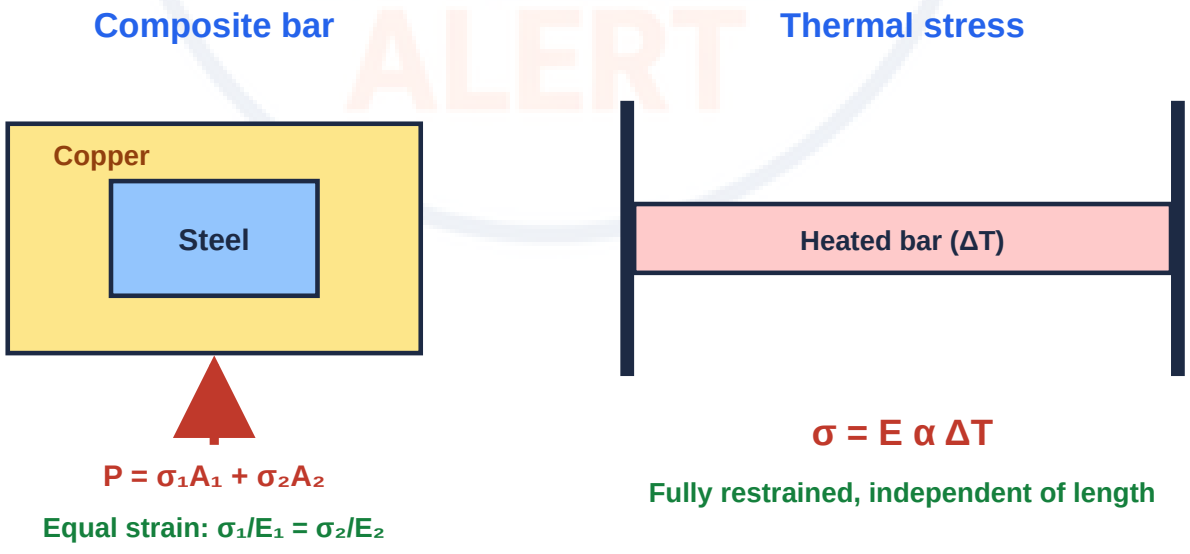



*Fig 3.4: Composite bar and thermal stress in a fully restrained bar*

## 2. Important Formulas & Facts

| Quantity / Topic | Formula / Fact |
|---|---|
| **Load balance** | $P = \sigma_1 A_1 + \sigma_2 A_2$ |
| **Equal strain** | $\sigma_1/E_1 = \sigma_2/E_2$ |
| **Modular ratio** | $m = E_1/E_2$ |
| **Equivalent E** | $E_{eq} = (E_1A_1 + E_2A_2)/(A_1 + A_2)$ |
| **Free thermal expansion** | $\delta = L\,\alpha\,\Delta T$ |
| **Thermal stress (full restraint)** | $\sigma = E\,\alpha\,\Delta T$ |
| **Partial restraint** | $\sigma = E(L\alpha\Delta T - \delta_g)/L$ |
| **Composite bar (thermal)** | $\sigma_1/E_1 + \sigma_2/E_2 = (\alpha_1 - \alpha_2)\Delta T$ |

## 3. Detailed Explanation

- **Statically indeterminate:** in a composite bar the unknown forces cannot be found from equilibrium alone; the compatibility (deformation) condition is also needed.
- **RCC column:** concrete and steel bars share the load; stress in steel = m × stress in concrete, with m ≈ 13–18.
- **Examples:** gaps between rails, expansion joints in bridges and loops in steam pipes are provided to avoid thermal stresses.
- **Why independent of length:** stress = E × prevented strain = E × αΔT; L cancels in the strain.
- **Derivation for composite bars:** the difference in free expansion $(\alpha_1 - \alpha_2)\Delta T \cdot L$ equals the sum of elastic strains: $\sigma_1/E_1 + \sigma_2/E_2 = (\alpha_1 - \alpha_2)\Delta T$, with $\sigma_1 A_1 = \sigma_2 A_2$.
- **Bolt and tube system:** tightening the nut puts the bolt in tension and the tube in compression; equal forces: $\sigma_b A_b = \sigma_t A_t$.

## 4. Comparison Table — Heating vs Cooling (restrained bar)

| Basis | Heating | Cooling |
|---|---|---|
| Tendency | Expand | Contract |
| Type of stress | Compressive | Tensile |
| Composite: higher-α material | Compression | Tension |

## 5. Solved Numerical Problems

**Example 1.** A 3 m steel rod is fixed at both ends and heated by 50°C. E = 200 GPa, $\alpha = 12 \times 10^{-6}$ /°C. Find the thermal stress.

**Solution:**

1. $\sigma = E\alpha\Delta T = 200000 \times 12 \times 10^{-6} \times 50$
2. $\sigma =$ **120 N/mm²** (compressive); length has no effect

**Example 2.** If the above rod is given a gap of 0.6 mm, find the stress.

**Solution:**

1. Free expansion = $3000 \times 12 \times 10^{-6} \times 50 = 1.8$ mm; prevented = $1.8 - 0.6 = 1.2$ mm
2. $\sigma = E \times 1.2/3000 =$ **80 N/mm²**

**Example 3.** A composite bar of steel (A = 1000 mm², E = 200 GPa) and copper (A = 2000 mm², E = 100 GPa) carries 400 kN compression. Find the stresses.

**Solution:**

1. $\sigma_s = (E_s/E_c)\sigma_c = 2\sigma_c$
2. $400000 = 2\sigma_c \times 1000 + \sigma_c \times 2000 = 4000\sigma_c$
3. $\sigma_c =$ **100 N/mm²**, $\sigma_s =$ **200 N/mm²**

**Example 4.** A steel (A = 500 mm²) – copper (A = 1000 mm²) composite bar is heated by 40°C. $E_s$ = 200, $E_c$ = 100 GPa; $\alpha_s = 12$, $\alpha_c = 18 \times 10^{-6}$ /°C. Find the stresses.

**Solution:**

1. Force balance: $\sigma_s \times 500 = \sigma_c \times 1000 \Rightarrow \sigma_s = 2\sigma_c$
2. $\sigma_s/200000 + \sigma_c/100000 = 6 \times 10^{-6} \times 40 = 2.4 \times 10^{-4}$
3. $2\sigma_c/200000 + \sigma_c/100000 = 2\sigma_c/100000 = 2.4 \times 10^{-4}$
4. $\sigma_c =$ **12 N/mm²** (compression), $\sigma_s =$ **24 N/mm²** (tension)

## 6. Frequently Asked Facts

- Thermal stress $\sigma = E\alpha\Delta T$ is independent of length and area.
- In a composite bar strain is equal; stresses are in the ratio of E.
- A restrained bar heated develops compressive stress.

## 7. Note (Confusing Facts)

- In a composite bar, strains (not stresses) are equal.
- Thermal stress arises only when expansion is prevented; a free bar has zero stress.
- On heating a composite bar, the lower-α material goes into tension.

**Remember:** One line: $\sigma_1/\sigma_2 = E_1/E_2$ (equal strain). Assertion–Reason: (A) Gaps are left between rails. (R) Prevented expansion in summer causes compressive stress. — Both true; R explains A.

# 30 Most Important One-Liner Facts — Composite Bars & Thermal Stresses

**1. What is equal in a composite bar?**

— Strain.

**2. Load in a composite bar?**

— $P = \sigma_1 A_1 + \sigma_2 A_2$.

**3. Modular ratio?**

— $E_1/E_2$.

**4. Which carries more stress?**

— Material with higher E.

**5. Thermal strain?**

— $\alpha\Delta T$.

**6. Free expansion?**

— $L\alpha\Delta T$.

**7. Thermal stress?**

— $E\alpha\Delta T$.

**8. Depends on length?**

— No.

**9. Heated restrained bar?**

— Compressive stress.

**10. Cooled restrained bar?**

— Tensile stress.

**11. Thermal stress in a free bar?**

— Zero.

**12. α of steel?**

— $\approx 12 \times 10^{-6}$ /°C.

**13. α of copper?**

— $\approx 17 \times 10^{-6}$ /°C.

**14. Composite bar is which type of problem?**

— Statically indeterminate.

**15. Stress in steel in RCC?**

— m × stress in concrete.

**16. Why gaps in rails?**

— To avoid thermal stress.

**17. Higher-α material on heating?**

— In compression.

**18. Stress with partial restraint?**

— $E(L\alpha\Delta T - \delta)/L$.

**19. Equivalent E?**

— $(E_1A_1 + E_2A_2)/(A_1 + A_2)$.

**20. α of aluminium?**

— $\approx 23 \times 10^{-6}$ /°C.

**21. Equivalent area?**

— $A_1 + mA_2$.

**22. Composite thermal equation?**

— $\sigma_1/E_1 + \sigma_2/E_2 = (\alpha_1 - \alpha_2)\Delta T$.

**23. Force balance (thermal)?**

— $\sigma_1 A_1 = \sigma_2 A_2$.

**24. Bolt in bolt–tube system?**

— Tension.

**25. Tube in bolt–tube system?**

— Compression.

**26. Flange bolts when hot pipe cools?**

— Tension.

**27. Modular ratio in RCC?**

— ≈ 13–18.

**28. Net force in composite bar (thermal, no load)?**

— Zero.

**29. Thermal stress independent of?**

— Length and area.

**30. Why expansion joints in bridges?**

— To avoid thermal stress.

# Practice MCQs — Composite Bars & Thermal Stresses

### Q1. Thermal stress in a fully restrained bar is:

(a) $E\alpha\Delta T$ (b) $L\alpha\Delta T$

(c) $E\alpha\Delta T/L$ (d) $\alpha\Delta T/E$

✔ **Answer: (a) $E\alpha\Delta T$**

**Explanation:** $\sigma = E \times \alpha\Delta T$.

---

### Q2. Thermal stress does not depend on:

(a) E (b) α

(c) Length (d) ΔT

✔ **Answer: (c) Length**

**Explanation:** L cancels out.

---

### Q3. In a composite bar both materials have equal:

(a) Stress (b) Strain

(c) Load (d) Area

✔ **Answer: (b) Strain**

**Explanation:** Both deform together.

---

### Q4. On heating a restrained bar, the stress is:

(a) Tensile (b) Compressive

(c) Shear (d) Zero

✔ **Answer: (b) Compressive**

**Explanation:** Prevented expansion pushes back.

---

### Q5. Modular ratio is:

(a) $E_1/E_2$ (b) $A_1/A_2$

(c) $\sigma_1 \times \sigma_2$ (d) $\alpha_1/\alpha_2$

✔ **Answer: (a) $E_1/E_2$**

**Explanation:** $m = E_1/E_2$.

---

### Q6. In a steel (200 GPa) – copper (100 GPa) composite bar, $\sigma_s/\sigma_c$ =

(a) 0.5 (b) 1

(c) 2 (d) 4

✔ **Answer: (c) 2**

**Explanation:** Equal strain gives stress ratio = E ratio.

---

### Q7. Thermal stress in a free bar on heating:

(a) Compressive
(b) Tensile
(c) Zero
(d) Shear

✔ **Answer: (c) Zero**

**Explanation:** Expansion is free.

---

### Q8. On heating a composite bar, the lower-α material is in:

(a) Compression
(b) Tension
(c) Zero stress
(d) Shear

✔ **Answer: (b) Tension**

**Explanation:** The higher-α material pulls it.

---

### Q9. E = 200 GPa, α = 12 × $10^{-6}$, ΔT = 100°C — thermal stress:

(a) 120 MPa
(b) 240 MPa
(c) 24 MPa
(d) 2400 MPa

✔ **Answer: (b) 240 MPa**

**Explanation:** 200000 × 12e-6 × 100 = 240.

---

### Q10. A composite bar problem is:

(a) Statically determinate
(b) Statically indeterminate
(c) Dynamic
(d) Unstable

✔ **Answer: (b) Statically indeterminate**

**Explanation:** Compatibility condition needed.

---

### Q11. A composite bar under temperature change with no external load:

(a) Both in tension
(b) One in tension, other in compression
(c) Both in compression
(d) No stress

✔ **Answer: (b) One in tension, other in compression**

**Explanation:** Internal forces balance.

---

### Q12. Tightening the nut in a bolt–tube system puts the tube in:

(a) Tension
(b) Compression
(c) Torsion
(d) Nothing

✔ **Answer: (b) Compression**

**Explanation:** Bolt in tension, tube in compression.

---

### Q13. Equivalent area of a composite bar (in material 1):

(a) $A_1 + A_2$
(b) $A_1 + mA_2$
(c) $mA_1 + A_2$
(d) $A_1A_2$

✔ **Answer: (b) $A_1 + mA_2$**

**Explanation:** $A_2$ is transformed by $m = E_2/E_1$.

---

**Q14. Correct equation for a composite bar on heating:**

(a) $\sigma_1/E_1 + \sigma_2/E_2 = (\alpha_1 - \alpha_2)\Delta T$ (b) $\sigma_1 E_1 = \sigma_2 E_2$

(c) $\sigma_1 + \sigma_2 = E\alpha\Delta T$ (d) $\sigma_1/\sigma_2 = \alpha_1/\alpha_2$

✔ **Answer: (a) $\sigma_1/E_1 + \sigma_2/E_2 = (\alpha_1 - \alpha_2)\Delta T$**

**Explanation:** Sum of strains = difference in expansion.

---

**Q15. Full restraint, E = 100 GPa, $\alpha = 20 \times 10^{-6}$, $\Delta T = 50$ — stress:**

(a) 50 MPa (b) 100 MPa

(c) 200 MPa (d) 10 MPa

✔ **Answer: (b) 100 MPa**

**Explanation:** 100000 × 20e-6 × 50 = 100.

---

## SSC JE / UPSSSC JE Previous Year Questions — Composite Bars & Thermal Stresses

**Q1. A bar fixed at both ends is heated. The stress induced is:** SSC JE (Previous Year)

(a) Tensile (b) Compressive

(c) Shear (d) Zero

✔ **Answer: (b) Compressive**

**Explanation:** Prevented expansion gives compression.

---

**Q2. Thermal stress is given by:** SSC JE (Previous Year)

(a) $\alpha\Delta T/E$ (b) $E\alpha\Delta T$

(c) $E\alpha\Delta T/L$ (d) $\alpha E/\Delta T$

✔ **Answer: (b) $E\alpha\Delta T$**

**Explanation:** $\sigma = E\alpha\Delta T$.

---

**Q3. In a composite bar, the material with higher E:** SSC JE (Previous Year)

(a) Takes less stress (b) Takes more stress

(c) Takes equal stress (d) Takes no load

✔ **Answer: (b) Takes more stress**

**Explanation:** $\sigma \propto E$.

---

**Q4. A bar fixed at both ends when cooled develops:** SSC JE (Previous Year)

(a) Compressive stress (b) Tensile stress

(c) Shear (d) Nothing

✔ **Answer: (b) Tensile stress**

**Explanation:** Contraction is prevented.

---

**Q5. Thermal strain is:**

SSC JE (Previous Year)

(a) $\alpha\Delta T$

(b) $E\alpha\Delta T$

(c) $L\alpha\Delta T$

(d) $\alpha\Delta T/E$

✔ **Answer: (a) $\alpha\Delta T$**

**Explanation:** $\varepsilon = \alpha\Delta T$.

UPSSSC ALERT

# Chapter 5: Torsion of Circular Shafts



## Torsion of Circular Shafts – Basic Concept

When a torque acts on a shaft about its axis, the shaft twists and **shear stress** develops in its cross-section. All power-transmitting shafts are subjected to torsion.

## 1. Key Points

- **Torsion equation:** $T/J = \tau/R = G\theta/L$, where T is torque, J polar moment of inertia, $\tau$ surface shear stress, R radius, G modulus of rigidity, $\theta$ angle of twist (radian), L length.
- **Polar moment of inertia:** solid shaft $J = \pi d^4/32$; hollow shaft $J = \pi(D^4 - d^4)/32$.
- **Torque capacity:** solid shaft $T = (\pi/16)\tau d^3$; hollow shaft $T = (\pi/16)\tau(D^4 - d^4)/D$.
- **Polar section modulus ($Z_p$):** $J/R$; $\pi d^3/16$ for a solid shaft. It measures torsional strength.
- **Power:** $P = 2\pi NT/60$ (W), N in rpm.
- **Stress distribution:** shear stress is zero at the centre and increases linearly with radius to a maximum at the surface.
- **Torsional rigidity:** GJ, the torque per unit twist per unit length. GJ/L is the torsional stiffness.
- **Hollow shaft is better:** for the same weight a hollow shaft is stronger and stiffer than a solid one, as material near the centre carries little stress.
- **Diameter for power:** find T from $P = 2\pi NT/60$, then $d = (16T/\pi\tau)^{1/3}$.
- **Hollow vs solid (same weight):** with inner diameter = ½ outer, a hollow shaft carries about 1.44 times the torque of a solid shaft of equal weight.
- **Combined bending and torsion:** equivalent torque $T_e = \sqrt{(M^2 + T^2)}$ and equivalent BM $M_e = \frac{1}{2}[M + \sqrt{(M^2 + T^2)}]$.

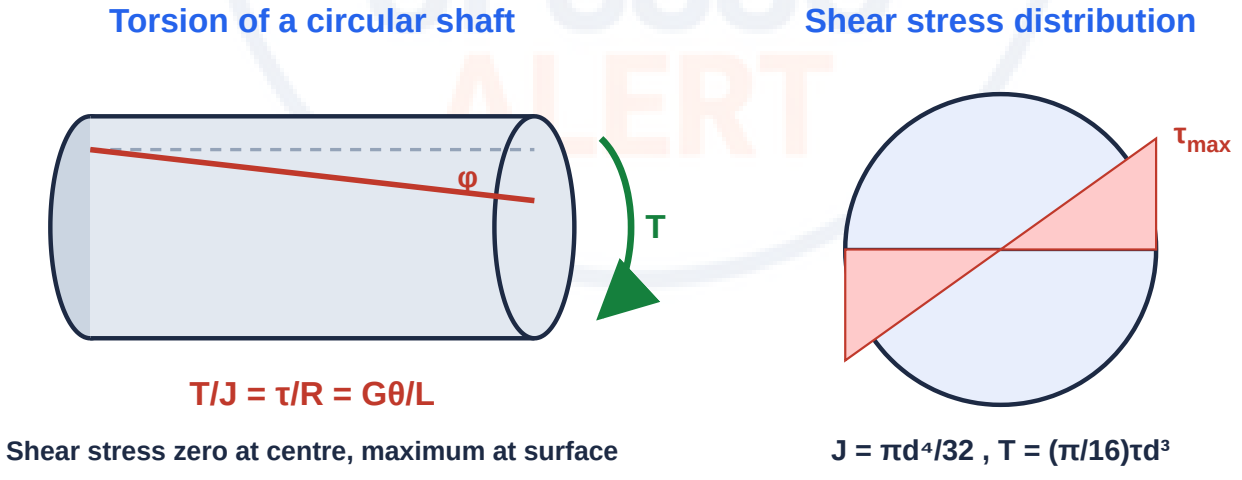



*Fig 3.5: Torsion of a circular shaft and shear stress distribution*

## 2. Important Formulas & Facts

| Quantity / Topic | Formula / Fact |
|---|---|
| **Torsion equation** | **$T/J = \tau/R = G\theta/L$** |
| **J (solid)** | **$\pi d^4/32$** |
| **J (hollow)** | **$\pi(D^4 - d^4)/32$** |
| **T (solid)** | **$(\pi/16)\,\tau\, d^3$** |
| **T (hollow)** | **$(\pi/16)\,\tau\,(D^4 - d^4)/D$** |
| **Power** | **$P = 2\pi NT / 60$** |
| **Angle of twist** | **$\theta = TL / GJ$** |
| **Shafts in series** | **Same T , $\theta = \theta_1 + \theta_2$** |
| **Shafts in parallel** | **Same $\theta$ , $T = T_1 + T_2$** |

## 3. Detailed Explanation

- **Assumptions:** material homogeneous and isotropic; plane sections remain plane; radii remain straight; stress within the elastic limit.
- **Series and parallel:** shafts in series carry the same torque and twists add; in parallel the twist is the same and torque divides.
- **Shaft design:** diameter is found from both strength ($\tau \le$ allowable) and stiffness ($\theta \le$ allowable, e.g. 1°/m), and the larger value is taken.
- **Doubling the diameter:** torque capacity becomes 8 times ($d^3$) and stiffness 16 times ($d^4$).
- **Close-coiled helical spring:** torsional stress in wire $\tau = 16WR/(\pi d^3)$; deflection $\delta = 64WR^3n/(Gd^4)$; stiffness $k = Gd^4/(64R^3n)$.
- **Spring combinations:** series $1/k = 1/k_1 + 1/k_2$; parallel $k = k_1 + k_2$.

## 4. Comparison Table — Solid vs Hollow Shaft

| Basis | Solid | Hollow |
|---|---|---|
| J | $\pi d^4/32$ | $\pi(D^4 - d^4)/32$ |
| Strength for same weight | Less | More |
| Material at centre | Under-used | Removed |
| Cost/manufacture | Cheaper, simpler | Costlier |

## 5. Solved Numerical Problems

**Example 1.** A 50 mm solid shaft has allowable shear stress 60 N/mm². Find the maximum torque.
**Solution:**

1. $T = (\pi/16) \times 60 \times 50^3 = (\pi/16) \times 60 \times 125000$
2. $T = 1.473 \times 10^6$ N·mm = **1.47 kN·m**

**Example 2.** A shaft transmits 1.5 kN·m at 200 rpm. Find the power.
**Solution:**

1. $P = 2\pi \times 200 \times 1500/60$ = **31.4 kW**

**Example 3.** For the shaft above (d = 50 mm, L = 2 m, G = 80 GPa) at 1.47 kN·m, find the angle of twist.
**Solution:**

1. $J = \pi \times 50^4/32 = 613{,}592$ mm⁴
2. $\theta = TL/GJ = 1.473 \times 10^6 \times 2000/(80000 \times 613592) = 0.060$ rad
3. $\theta$ = **3.44°**

**Example 4.** A solid shaft transmits 30 kW at 150 rpm; $\tau$ = 60 N/mm². Find the diameter.
**Solution:**

1. $T = 30000 \times 60/(2\pi \times 150) = 1910$ N·m $= 1.91 \times 10^6$ N·mm
2. $d^3 = 16T/(\pi\tau) = 16 \times 1.91 \times 10^6/(\pi \times 60) = 162{,}100$
3. d = **54.5 mm ≈ 55 mm**

**Example 5.** A shaft has M = 3 kN·m and T = 4 kN·m. Find equivalent torque and BM.
**Solution:**

1. $T_e = \sqrt{(9 + 16)}$ = **5 kN·m**
2. $M_e = \frac{1}{2}(3 + 5)$ = **4 kN·m**

## 6. Frequently Asked Facts

- $T/J = \tau/R = G\theta/L$ — the whole chapter is built on this.
- Solid shaft: $T = (\pi/16)\tau d^3$; $P = 2\pi NT/60$.
- A hollow shaft is stronger for the same weight.

## 7. Note (Confusing Facts)

- Shear stress is zero at the centre and maximum at the surface — not the reverse.
- $\theta$ is in radians; multiply by $180/\pi$ for degrees.
- Diameter doubled → strength 8 times, stiffness 16 times.

**Remember:** One line: GJ = torsional rigidity; J/R = polar section modulus. Assertion–Reason: (A) A hollow shaft is stronger than a solid one of the same weight. (R) Material near the centre is lightly

stressed. — Both true; R explains A.

## 30 Most Important One-Liner Facts — Torsion of Circular Shafts

**1. Torsion equation?**

— $T/J = \tau/R = G\theta/L$.

**2. J of solid shaft?**

— $\pi d^4/32$.

**3. J of hollow shaft?**

— $\pi(D^4 - d^4)/32$.

**4. Torque of solid shaft?**

— $(\pi/16)\tau d^3$.

**5. Power formula?**

— $2\pi NT/60$.

**6. Angle of twist?**

— $TL/GJ$.

**7. Torsional rigidity?**

— $GJ$.

**8. Polar section modulus (solid)?**

— $\pi d^3/16$.

**9. Shear stress at centre?**

— Zero.

**10. Maximum shear stress at?**

— Outer surface.

**11. Stress distribution?**

— Linear with radius.

**12. Diameter doubled — strength?**

— 8 times.

**13. Diameter doubled — stiffness?**

— 16 times.

**14. Same in series shafts?**

— Torque.

**15. Same in parallel shafts?**

— Angle of twist.

**16. Better shaft for same weight?**

— Hollow.

**17. Unit of θ in formula?**

— Radian.

**18. Type of stress in torsion?**

— Shear.

**19. G in torsion?**

— Modulus of rigidity.

**20. Two design criteria for shafts?**

— Strength and stiffness.

**21. Diameter from power?**

— $(16T/\pi\tau)^{1/3}$.

**22. Equivalent torque?**

— $\sqrt{(M^2 + T^2)}$.

**23. Equivalent BM?**

— $\frac{1}{2}[M + \sqrt{(M^2 + T^2)}]$.

**24. Stress in spring wire?**

— $16WR/\pi d^3$.

**25. Spring deflection?**

— $64WR^3n/Gd^4$.

**26. Series spring stiffness?**

— $1/k = 1/k_1 + 1/k_2$.

**27. Parallel spring stiffness?**

— $k_1 + k_2$.

**28. Stress in close-coiled spring?**

— Torsional (shear).

**29. 1 kW equals?**

— 1000 W.

**30. G of steel?**

— ≈ 80 GPa.

# Practice MCQs — Torsion of Circular Shafts

### Q1. The torsion equation is:

(a) $T/J = \tau/R = G\theta/L$

(b) $M/I = \sigma/y = E/R$

(c) $T/I = \tau/R = E\theta/L$

(d) $P/A = \sigma$

✔ **Answer: (a) $T/J = \tau/R = G\theta/L$**

**Explanation:** Torsion equation.

---

### Q2. Polar moment of inertia of a solid shaft:

(a) $\pi d^4/64$

(b) $\pi d^4/32$

(c) $\pi d^3/16$

(d) $\pi d^3/32$

✔ **Answer: (b) $\pi d^4/32$**

**Explanation:** $J = 2I = \pi d^4/32$.

---

### Q3. Torque capacity of a solid shaft:

(a) $(\pi/16)\tau d^3$

(b) $(\pi/32)\tau d^3$

(c) $(\pi/16)\tau d^4$

(d) $(\pi/32)\tau d^4$

✔ **Answer: (a) $(\pi/16)\tau d^3$**

**Explanation:** $T = \tau \times J/R$.

---

### Q4. Shear stress at the centre of a shaft:

(a) Maximum

(b) Zero

(c) Average

(d) Infinite

✔ **Answer: (b) Zero**

**Explanation:** $\tau \propto r$.

---

### Q5. Doubling the diameter, the strength of a shaft becomes:

(a) 2 times

(b) 4 times

(c) 8 times

(d) 16 times

✔ **Answer: (c) 8 times**

**Explanation:** $T \propto d^3$.

---

### Q6. Torsional rigidity is:

(a) GJ

(b) EI

(c) GA

(d) EJ

✔ **Answer: (a) GJ**

**Explanation:** GJ.

---

### Q7. Power transmitted by a shaft:

(a) $2\pi NT/60$ (b) $\pi NT/60$

(c) $2\pi NT$ (d) $NT/60$

✔ **Answer: (a) $2\pi NT/60$**

**Explanation:** $P = T\omega$.

---

### Q8. In shafts connected in series, the same is:

(a) Angle of twist (b) Torque

(c) Diameter (d) Stress

✔ **Answer: (b) Torque**

**Explanation:** The same torque passes through both.

---

### Q9. A hollow shaft compared to a solid one of same weight is:

(a) Weaker (b) Stronger

(c) Equal (d) Cannot say

✔ **Answer: (b) Stronger**

**Explanation:** Material is used where stress is high.

---

### Q10. Stress produced in torsion:

(a) Tensile (b) Compressive

(c) Shear (d) Bending

✔ **Answer: (c) Shear**

**Explanation:** Shear stress.

---

### Q11. Equivalent torque under combined bending and torsion:

(a) $M + T$ (b) $\sqrt{(M^2 + T^2)}$

(c) $\frac{1}{2}(M + T)$ (d) $M^2 + T^2$

✔ **Answer: (b) $\sqrt{(M^2 + T^2)}$**

**Explanation:** $T_e = \sqrt{(M^2 + T^2)}$.

---

### Q12. Stiffness of two springs in series:

(a) $k_1 + k_2$ (b) $k_1k_2/(k_1 + k_2)$

(c) $k_1 - k_2$ (d) $\sqrt{(k_1k_2)}$

✔ **Answer: (b) $k_1k_2/(k_1 + k_2)$**

**Explanation:** $1/k = 1/k_1 + 1/k_2$.

---

### Q13. Stress in the wire of a close-coiled spring:

(a) Bending (b) Torsional shear

(c) Tensile (d) Compressive

✔ **Answer: (b) Torsional shear**

**Explanation:** The wire is in torsion.

---

### Q14. M = 3, T = 4 kN·m — equivalent BM:

(a) 3.5
(b) 4
(c) 5
(d) 7

✔ **Answer: (b) 4**

**Explanation:** ½(3 + 5) = 4.

---

### Q15. Spring deflection varies with wire diameter as:

(a) d
(b) 1/d
(c) $1/d^3$
(d) $1/d^4$

✔ **Answer: (d) $1/d^4$**

**Explanation:** $\delta \propto 1/d^4$.

---

## SSC JE / UPSSSC JE Previous Year Questions — Torsion of Circular Shafts

### Q1. Polar moment of inertia of a hollow shaft:

SSC JE (Previous Year)

(a) $\pi(D^4 - d^4)/32$
(b) $\pi(D^4 - d^4)/64$
(c) $\pi(D^3 - d^3)/16$
(d) $\pi(D^2 - d^2)/4$

✔ **Answer: (a) $\pi(D^4 - d^4)/32$**

**Explanation:** $J = \pi(D^4 - d^4)/32$.

---

### Q2. Maximum shear stress in a shaft occurs at:

SSC JE (Previous Year)

(a) Centre
(b) Outer surface
(c) Half radius
(d) Everywhere equal

✔ **Answer: (b) Outer surface**

**Explanation:** $\tau = Tr/J$, maximum at r = R.

---

### Q3. Polar section modulus of a solid shaft:

SSC JE (Previous Year)

(a) $\pi d^3/16$
(b) $\pi d^3/32$
(c) $\pi d^4/32$
(d) $\pi d^2/4$

✔ **Answer: (a) $\pi d^3/16$**

**Explanation:** $J/R = \pi d^3/16$.

---

### Q4. For the same weight, compared to a solid shaft, a hollow shaft is:

SSC JE (Previous Year)

(a) Weaker
(b) Stronger
(c) Equal
(d) Less stiff

✔ **Answer: (b) Stronger**

**Explanation:** Material at outer region.

---

**Q5. In T/J = τ/R, J is:**

SSC JE (Previous Year)

(a) Moment of inertia

(b) Polar moment of inertia

(c) Section modulus

(d) Joule

✔ **Answer: (b) Polar moment of inertia**

**Explanation:** Polar moment of inertia.

UPSSSC ALERT

# Chapter 6: Thin-walled Pressure Vessels



## Thin-walled Pressure Vessels – Basic Concept

A cylindrical or spherical vessel whose wall thickness is less than 1/20 of its diameter ($t < d/20$) is a **thin vessel**. Boiler drums, gas cylinders, pipes and tanks are examples. Internal pressure produces hoop and longitudinal stresses in the wall.

## 1. Key Points

- **Hoop (circumferential) stress:** $\sigma_h = pd/2t$; it acts on the longitudinal section and tends to split the cylinder along its length.
- **Longitudinal stress:** $\sigma_L = pd/4t = \sigma_h/2$; it tends to separate the cylinder into two parts around the circumference.
- **Thin sphere:** $\sigma = pd/4t$ in all directions; so for the same pressure and diameter a sphere needs half the thickness of a cylinder.
- **Maximum shear stress (cylinder):** $\tau_{max} = (\sigma_h - \sigma_L)/2 = pd/8t$ (in-plane).
- **Design:** thickness from hoop stress: $t = pd/(2\sigma_t\eta_L)$, where $\eta_L$ is the efficiency of the longitudinal joint. Longitudinal stress uses the circumferential joint efficiency $\eta_c$.
- **Volumetric strain (cylinder):** $\delta V/V = (pd/4tE)(5 - 4\mu)$; for a sphere $\delta V/V = (3pd/4tE)(1 - \mu)$.
- **Radial stress:** it is neglected in thin vessels; in thick cylinders it is found by Lamé's equations.
- **Thick cylinder:** $t > d/20$; hoop stress is maximum at the inner surface and radial stress varies from $-p$ inside to zero outside (Lamé).
- **Change in diameter and length:** $\delta d/d = (pd/4tE)(2 - \mu)$; $\delta L/L = (pd/4tE)(1 - 2\mu)$.
- **Wire-wound cylinder:** winding wire under initial tension produces compressive hoop stress, increasing pressure capacity (gun barrels).

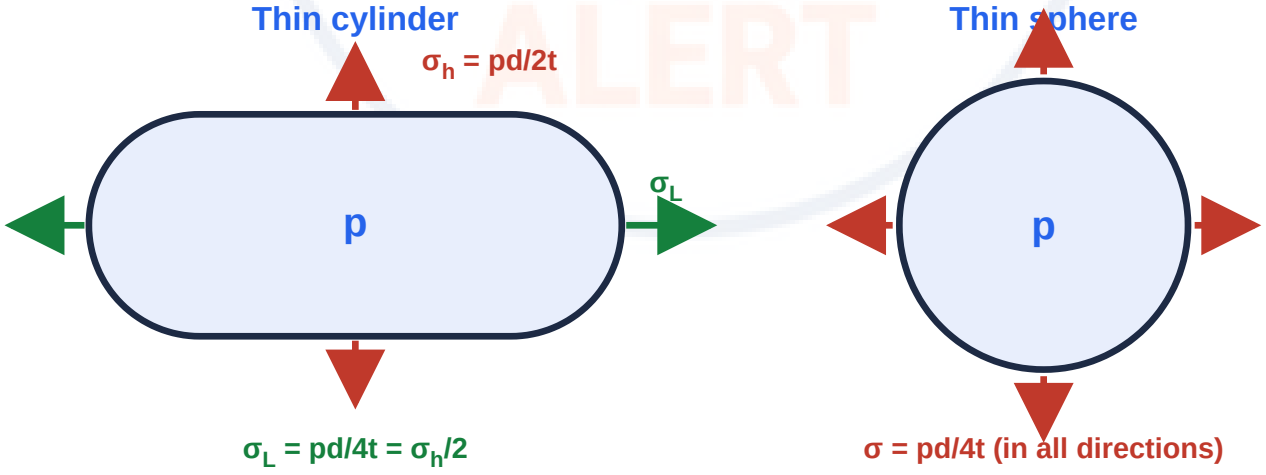



*Fig 3.6: Stresses in a thin cylinder and a thin sphere*

## 2. Important Formulas & Facts

| Quantity / Topic | Formula / Fact |
|---|---|
| **Hoop stress** | **$\sigma_h = pd / 2t$** |
| **Longitudinal stress** | **$\sigma_L = pd / 4t$** |
| **Ratio** | **$\sigma_h : \sigma_L = 2 : 1$** |
| **Thin sphere** | **$\sigma = pd / 4t$** |
| **Max in-plane shear** | **$\tau = pd / 8t$** |
| **With joint efficiency** | **$\sigma_h = pd/(2t\eta_L)$ , $\sigma_L = pd/(4t\eta_c)$** |
| **Volumetric strain of cylinder** | **$(pd/4tE)(5 - 4\mu)$** |
| **Thin vessel condition** | **$t < d/20$** |

## 3. Detailed Explanation

- **Derivation of hoop stress:** bursting force on half cylinder = p × d × L, resisted by two walls: $2 \times \sigma_h \times t \times L$. Equating gives $\sigma_h = pd/2t$.
- **Derivation of longitudinal stress:** end force = $p \times \pi d^2/4$, resisting area ≈ $\pi d \times t$; so $\sigma_L = pd/4t$.
- **Why a cylinder splits lengthwise:** the hoop stress is double, so the crack runs longitudinally; hence the longitudinal seam is made stronger.
- **Hemispherical ends of gas cylinders:** spherical ends have lower stress, so a thinner wall suffices.
- **Volumetric strain of a sphere:** $\delta V/V = 3 \times$ diametral strain $= (3pd/4tE)(1 - \mu)$.
- **Codes:** in India boilers follow IBR and vessels IS 2825; a corrosion allowance is added to the thickness.

## 4. Comparison Table — Thin Cylinder vs Thin Sphere

| Basis | Cylinder | Sphere |
|---|---|---|
| Maximum stress | pd/2t (hoop) | pd/4t |
| Stress directions | Hoop ≠ longitudinal | Equal in all directions |
| Thickness needed (same p, d) | t | t/2 |
| Examples | Boiler drum, pipes | Gas storage tanks |

## 5. Solved Numerical Problems

**Example 1.** A cylinder of 1 m diameter and 10 mm thickness has internal pressure 2 N/mm². Find the hoop and longitudinal stresses.

**Solution:**

1. $\sigma_h = pd/2t = 2 \times 1000/20 =$ **100 N/mm²**
2. $\sigma_L = pd/4t =$ **50 N/mm²**

**Example 2.** A 1.5 m boiler works at 1.2 N/mm²; allowable stress 90 N/mm², longitudinal joint efficiency 80%. Find the thickness.

**Solution:**

1. $t = pd/(2\sigma_t\eta) = 1.2 \times 1500/(2 \times 90 \times 0.8)$
2. $t = 1800/144 =$ **12.5 mm**

**Example 3.** A 2 m thin sphere withstands 1 N/mm²; allowable stress 100 N/mm². Find the thickness.

**Solution:**

1. $t = pd/4\sigma = 1 \times 2000/400 =$ **5 mm**

**Example 4.** A cylinder (d = 800 mm, t = 8 mm) has p = 1.5 N/mm²; E = 200 GPa, μ = 0.3. Find volumetric strain.

**Solution:**

1. $pd/4tE = 1.5 \times 800/(4 \times 8 \times 200000) = 1.875 \times 10^{-4}$
2. $\delta V/V = 1.875 \times 10^{-4} \times (5 - 1.2) =$ **$7.125 \times 10^{-4}$**

**Example 5.** A cylinder of 1.2 m diameter and 12 mm thickness has allowable hoop stress 100 N/mm². Find the maximum pressure.

**Solution:**

1. $p = 2t\sigma/d = 2 \times 12 \times 100/1200 =$ **2 N/mm²**

## 6. Frequently Asked Facts

- $\sigma_h = pd/2t$, $\sigma_L = pd/4t$; ratio 2 : 1.
- Thin sphere $\sigma = pd/4t$; half the cylinder's thickness is enough.
- Thin vessel: $t < d/20$.

## 7. Note (Confusing Facts)

- Hoop stress acts on the longitudinal section, so the cylinder bursts along its length.
- In thickness design, longitudinal joint efficiency is used with hoop stress.
- Radial stress in thin vessels is negligible and taken as zero.

**Remember:** One line: hoop = 2 × longitudinal. Assertion–Reason: (A) A boiler's longitudinal seam is a butt joint. (R) Hoop stress is greater. — Both true; R explains A.

## 30 Most Important One-Liner Facts — Thin-walled Pressure Vessels

**1. Condition for thin vessel?**

— $t < d/20$.

**2. Hoop stress?**

— $pd/2t$.

**3. Longitudinal stress?**

— $pd/4t$.

**4. Ratio hoop : longitudinal?**

— 2 : 1.

**5. Stress in thin sphere?**

— pd/4t.

**6. Cylinder bursts along?**

— Its length.

**7. Max in-plane shear?**

— pd/8t.

**8. Radial stress in thin vessel?**

— Negligible.

**9. Equations for thick cylinder?**

— Lamé's.

**10. Sphere thickness for same p, d?**

— Half of cylinder.

**11. Another name for hoop stress?**

— Circumferential stress.

**12. Hoop stress acts on?**

— Longitudinal section.

**13. Thickness design formula?**

— $t = pd/(2\sigma\eta_L)$.

**14. Efficiency used for longitudinal stress?**

— Circumferential joint.

**15. Volumetric strain of cylinder?**

— $(pd/4tE)(5 - 4\mu)$.

**16. Volumetric strain of sphere?**

— $(3pd/4tE)(1 - \mu)$.

**17. Examples of thin vessels?**

— Boiler, gas cylinder, pipes.

**18. Hoop stress tends to?**

— Split cylinder lengthwise.

**19. Longitudinal stress tends to?**

— Separate cylinder circumferentially.

**20. Why spherical storage tanks?**

— Lower stress, less material.

**21. Thick cylinder condition?**

— $t > d/20$.

**22. Max hoop stress in thick cylinder?**

— Inner surface.

**23. Diametral strain?**

— $(pd/4tE)(2 - \mu)$.

**24. Longitudinal strain?**

— $(pd/4tE)(1 - 2\mu)$.

**25. Benefit of wire winding?**

— Higher pressure capacity.

**26. Boiler design code?**

— IBR.

**27. Extra allowance in thickness?**

— Corrosion allowance.

**28. Max pressure from σ?**

— $p = 2t\sigma/d$.

**29. Diametral strain of sphere?**

— $(pd/4tE)(1 - \mu)$.

**30. Principal stresses in thin vessel?**

— Hoop and longitudinal.

## Practice MCQs — Thin-walled Pressure Vessels

### Q1. Hoop stress in a thin cylinder:

(a) pd/t

(b) pd/2t

(c) pd/4t

(d) pd/8t

✔ **Answer: (b) pd/2t**

**Explanation:** $\sigma_h = pd/2t$.

### Q2. Longitudinal stress in a thin cylinder:

(a) pd/2t
(b) pd/4t
(c) pd/t
(d) 2pd/t

✔ **Answer: (b) pd/4t**

**Explanation:** $\sigma_L$ = pd/4t.

---

### Q3. Ratio of hoop to longitudinal stress:

(a) 1 : 1
(b) 1 : 2
(c) 2 : 1
(d) 4 : 1

✔ **Answer: (c) 2 : 1**

**Explanation:** pd/2t ÷ pd/4t = 2.

---

### Q4. Stress in a thin sphere:

(a) pd/2t
(b) pd/4t
(c) pd/8t
(d) pd/t

✔ **Answer: (b) pd/4t**

**Explanation:** pd/4t in all directions.

---

### Q5. A vessel is considered thin when:

(a) t < d/20
(b) t > d/20
(c) t = d/10
(d) t < d/4

✔ **Answer: (a) t < d/20**

**Explanation:** t/d < 1/20.

---

### Q6. A thin cylinder under excess pressure bursts:

(a) Circumferentially
(b) Along its length
(c) At the ends
(d) Anywhere

✔ **Answer: (b) Along its length**

**Explanation:** Hoop stress is higher.

---

### Q7. Max in-plane shear stress in a thin cylinder:

(a) pd/4t
(b) pd/8t
(c) pd/2t
(d) pd/16t

✔ **Answer: (b) pd/8t**

**Explanation:** (pd/2t − pd/4t)/2.

---

### Q8. For the same p and d, thickness of a thin sphere is ____ that of a cylinder:

(a) Double
(b) Half
(c) Equal
(d) One-fourth

✔ **Answer: (b) Half**

**Explanation:** Sphere stress is pd/4t.

---

**Q9. d = 1 m, t = 10 mm, p = 2 N/mm² — hoop stress:**

(a) 50 MPa
(b) 100 MPa
(c) 200 MPa
(d) 25 MPa

✔ **Answer: (b) 100 MPa**

**Explanation:** 2 × 1000/20 = 100.

---

**Q10. Thick cylinders are analysed by:**

(a) Lamé's equations
(b) Euler's formula
(c) Unwin's formula
(d) Bernoulli's equation

✔ **Answer: (a) Lamé's equations**

**Explanation:** Lamé.

---

**Q11. Maximum hoop stress in a thick cylinder occurs at:**

(a) Outer surface
(b) Inner surface
(c) Mid-thickness
(d) Uniform

✔ **Answer: (b) Inner surface**

**Explanation:** Lamé.

---

**Q12. Diametral strain of a thin cylinder:**

(a) (pd/4tE)(2 − μ)
(b) (pd/4tE)(1 − 2μ)
(c) (pd/4tE)(5 − 4μ)
(d) pd/2tE

✔ **Answer: (a) (pd/4tE)(2 − μ)**

**Explanation:** $\sigma_h/E - \mu\sigma_L/E$.

---

**Q13. Winding wire on a cylinder produces:**

(a) Compressive hoop stress
(b) Tensile hoop stress
(c) Shear
(d) Nothing

✔ **Answer: (a) Compressive hoop stress**

**Explanation:** Initial compression.

---

**Q14. d = 1.2 m, t = 12 mm, σ = 100 MPa — maximum pressure:**

(a) 1 MPa
(b) 2 MPa
(c) 4 MPa
(d) 0.5 MPa

✔ **Answer: (b) 2 MPa**

**Explanation:** 2tσ/d = 2.

---

**Q15. Boilers in India are designed as per:**

(a) IBR
(b) ASME
(c) BIS 456
(d) ISO 9001

✔ **Answer: (a) IBR**

**Explanation:** Indian Boiler Regulations.

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# SSC JE / UPSSSC JE Previous Year Questions — Thin-walled Pressure Vessels

**Q1. In a thin cylinder, hoop stress is ____ the longitudinal stress:** SSC JE (Previous Year)

(a) Half
(b) Equal
(c) Twice
(d) Four times

✔ **Answer: (c) Twice**

**Explanation:** 2 : 1.

---

**Q2. Stress in a thin spherical shell:** SSC JE (Previous Year)

(a) pd/2t
(b) pd/4t
(c) pd/t
(d) 2pd/t

✔ **Answer: (b) pd/4t**

**Explanation:** pd/4t.

---

**Q3. Thickness of a boiler shell is decided by:** SSC JE (Previous Year)

(a) Longitudinal stress
(b) Hoop stress
(c) Radial stress
(d) Shear stress

✔ **Answer: (b) Hoop stress**

**Explanation:** Hoop stress is maximum.

---

**Q4. Longitudinal stress in a thin cylinder:** SSC JE (Previous Year)

(a) pd/2t
(b) pd/4t
(c) 2pd/t
(d) pd/t

✔ **Answer: (b) pd/4t**

**Explanation:** pd/4t.

---

**Q5. Volumetric strain of a thin cylinder:** SSC JE (Previous Year)

(a) (pd/4tE)(5 − 4μ)
(b) (pd/4tE)(5 − 2μ)
(c) (pd/2tE)(1 − μ)
(d) (3pd/4tE)(1 − μ)

✔ **Answer: (a) (pd/4tE)(5 − 4μ)**

**Explanation:** 2 × diametral + longitudinal strain.

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